

Plot MCMC Chain Trace and Features Posterior Density

```
par(mar=c(1,1,1,1))
plot(samples1)
```

Descriptive Statistics of model1

```
summary(samples1)
```

Show Alpha and Betas to the corresponding Features

```
sum <- summary(samples1)
rownames(sum$statistics) <- names
rownames(sum$quantiles) <- names
sum$statistics <- round(sum$statistics,3)
sum$quantiles <- round(sum$quantiles,3)
sum
```

Check Convergence

```
gelman.diag(samples1)
```

Compile results

```
ESS1 <- effectiveSize(samples1)
out1 <- summary(samples1)$quantiles
rownames(out1)<-names

ESS1
```

Compute DIC & WAIC

DIC - WAIC : For Model Selection The Lower the better

```
# DIC
dic1 <- dic.samples(model1,n.iter=n.iter)

# WAIC
waic1 <- coda.samples(model1, variable.names=c("like"), n.iter=n.iter)
like1 <- waic1[[1]]
fbar1 <- colMeans(like1)
P1 <- sum(apply(log(like1),2,var))
WAIC1 <- -2*sum(log(fbar1))+2*P1
```

Plot Posterior Density Individually

```
beta <- NULL
for(l in 1:n.chains){
  beta <- rbind(beta,samples1[[l]])
}
colnames(beta) <- names
for(j in 1:17){
  hist(beta[,j],xlab=expression(beta[j]),ylab="Posterior density",
  breaks=100,main=names[j])
}
```

For Feature Selection

```
library(knitr)
Inc_Prob <- apply(beta!=0,2,mean)
Q <- t(apply(beta,2,quantile,c(0.5,0.05,0.95)))
out <- cbind(Inc_Prob,Q)
kable(round(out,2))
```